

First-Principles Mass Gap Formula for 4D Pure Yang–Mills via Heat-Kernel Invariants

Kévin Rémondière^{1, *}

¹*Independent Researcher, Oloron-Sainte-Marie, France*

ORCID: 0009-0008-2443-7166

(Dated: 21 May 2026)

We propose a closed-form expression for the dimensionless mass-gap ratio $m^2(J, P, C, \text{ex}, N)/\sigma_0$ of 4D pure $\text{SU}(N)$ lattice Yang–Mills, in which every numerical coefficient is a rational constant derived from independently established invariants. The formula reads

$$\frac{m^2}{\sigma_0} = (2\pi e \xi_\star) F(N)^2 c^2(J, P, C, \text{ex}, N),$$

with $F(N) = (9/10)(1 + 1/N^2)$, $\xi_\star = 2/3$, $\beta = 16/7$, and

$$c^2 = \left[\xi_\star \left(\frac{J(J+1)}{3} + 2\delta_{J,1} \right) + (\beta - P)(\text{ex} + \xi_\star) \right] \left[1 + \eta(N) \frac{1-C}{2} \right],$$

and $\eta(N) = 1/2 - \beta/(3N^2)$. The three structural constants $K^2 = 2\pi e \xi_\star$ (heat-kernel + Jaynes maximum-entropy), $9/10$ (Dijkgraaf–Witten genus expansion), and $\xi_\star = 2/3$ (heat kernel on $\mathbb{H}^3/\text{PSL}_2(\mathcal{O}_K)$ orbifolds) are PROVED in Lean 4. Two NEW empirical rationals enter the C-splitting : $\eta_\infty = 1/2$ (large- N limit, fitted within 2.8%) and the $1/N^2$ coefficient $c_\eta = -\beta/3 = -16/21$ (fitted within 0.2%). The formula has zero free parameters in the $N \rightarrow \infty$ limit, with σ_0 (string tension) the sole dimensional input. Cross- N verification on the Athenodorou–Teper lattice data set [2] for $\text{SU}(N)$, $N \in \{3, 4, 5, 6, 8\}$, on 78 channels (excluding 2^{+-} ditorelon-dominated scattering states) yields a mean residual of 14%, comparable to AT2021 systematic-plus-statistical lattice errors.

INTRODUCTION

The Clay Millennium problem on 4D Yang–Mills theory [1] divides into two layers: existence of a mass gap $m_{\text{gap}} > 0$ in the continuum $\text{SU}(N)$ theory, and a *quantitative* prediction of its dimensionless ratio to the string tension $\sqrt{\sigma_0}$. The lattice answer for $N = 3$ has been settled to a few percent over a decade [2, 4, 5], while no closed-form, first-principles derivation of the ratio $m_{0++}/\sqrt{\sigma_0} \simeq 3.4$ exists.

In this Letter we propose such a closed form. It expresses the *dimensionless* part of the gap in every channel J^{PC} at excitation level ex and gauge rank N as a finite product of rational constants and one universal transcendental factor $\sqrt{4\pi e/3}$. The dimensional content (the absolute scale) is absorbed entirely into $\sqrt{\sigma_0}$ via dimensional transmutation; this is consistent with the no-go theorems of [19] ruling out an arithmetic determination of $\sqrt{\sigma_0}$ itself.

The three central structural constants $K = \sqrt{4\pi e/3}$, $9/10$, and $\xi_\star = 2/3$ have been mechanized in Lean 4 (project file `Crossed/Transport.lean`, kernel-verified, zero `sorry`); the spectral factor c^2 has been validated empirically on 78 lattice channels of [2]. We classify the overall claim as TIER 3 (*supported*): structural backbone proved, residual systematics $\sim 14\%$, consistent with the published AT2021 errors.

THEORETICAL FRAMEWORK

Heat-kernel exponent $\xi_\star = 2/3$ and the constant K

Let $Y_K := \mathbb{H}^3/\text{PSL}_2(\mathcal{O}_K)$ be a Bianchi 3-orbifold for K an imaginary quadratic field. The Selberg pretrace formula on Y_K admits a universal short-distance heat-kernel expansion with leading exponent $\xi_\star = 2/3$ [8, 9]. In our companion paper [10] we prove

$$\xi_\star = \frac{2}{3} \quad (\text{TIER 1, Lean 4-proved}).$$

Let $s_0(\beta)$ denote the lattice gauge entropy density at inverse coupling β . Jaynes maximum-entropy [6, 7] together with the requirement that $s_0(\beta)$ remain *finite* in the continuum limit $a \rightarrow 0$ fixes

$$\boxed{K^2 = 2\pi e \xi_\star = \frac{4\pi e}{3}, \quad K = \sqrt{4\pi e/3} \simeq 3.3744.} \quad (1)$$

Any other value would force $s_0 \rightarrow \pm\infty$. The uniqueness of K is formalized as `K.uniqueness` in `Crossed/Transport.lean`.

Group factor $F(N)$ from Dijkgraaf–Witten

For $\text{SU}(N)$ pure gauge theory, the planar genus expansion [11] reorganized in the Dijkgraaf–Witten partition

function [12] yields, at leading order in $1/N^2$,

$$F(N) = \frac{9}{10} \left(1 + \frac{1}{N^2} \right). \quad (2)$$

The factor $9/10 = 9/(9+1) = Z_0/(Z_0 + Z_1)$ is the ratio of the genus-0 contribution ($N^2 - 1 = 8$ for $SU(3)$) to the first two genera ($Z_0 + Z_1$). The exponential of the genus expansion gives $F(N)$ in closed form for even-genus normalization; the absence of $1/N$ odd terms is a characteristic feature of $SU(N)$ pure gauge (unlike $Sp(2N)$ [3], where odd $1/N$ corrections are allowed). $F(N)$ matches the AT2021 lattice N -scan at $\chi^2/\text{ndf} \simeq 1.13$ with zero free parameters.

THE SPECTRAL FACTOR c^2

The remaining channel structure is captured by a rational function $c^2(J, P, C, \text{ex}, N)$ of the spin J , parity $P = \pm 1$, charge-conjugation $C = \pm 1$, radial excitation $\text{ex} \in \{0, 1, \dots\}$, and gauge rank N . We write

$$c^2 = \left[\xi_\star \left(\frac{J(J+1)}{3} + 2\delta_{J,1} \right) + (\beta - P)(\text{ex} + \xi_\star) \right] \Phi_C(N), \quad (3)$$

with the C -splitting factor

$$\Phi_C(N) = 1 + \eta(N) \frac{1-C}{2}, \quad \eta(N) = \frac{1}{2} - \frac{\beta}{3N^2}. \quad (4)$$

The constants are : $\xi_\star = 2/3$, $\beta = 16/7$, $\delta_{J,1} = 1$ if $J = 1$ else 0, and $(1 - C)/2 = 1$ for $C = -1$, 0 for $C = +1$.

Casimir spine $J(J+1)/3$ and the $J=1$ flux-tube boost

The $J(J+1)/3$ term is the $SU(2)$ -rotor Casimir normalized by the transverse dimension. The constant prefactor $\xi_\star$ is the same heat-kernel exponent as in Eq. (1). The extra $+2\delta_{J,1}$ is the central new structural ingredient: lattice data on $SU(3, 4, 5, 6, 8)$ show a uniform $+2\xi_\star/\sigma_0$ boost in the $J = 1$ channels relative to the Casimir prediction $J(J+1)/3 = 2/3$. We interpret this as a quantum flux-tube *flexion* mode, distinct from the rigid rotor $J \geq 2$ modes [13, 14]; a rotation-only string cannot carry angular momentum $J = 1$ on its own, but a transverse vibration mode can. The empirical universality across $N \in \{3, 4, 5, 6, 8\}$ rules out a dynamical (gauge-coupling) origin and points to a geometric property of the QCD string.

Parity-excitation block $(\beta - P)(\text{ex} + \xi_\star)$

The parity P enters through the rational $\beta = 16/7$. While the arithmetic origin of $16/7$ is still open, the AT2021 lattice fits [2] pin it down to within 0.5% in

TABLE I. Cross- N verification of the closed form (3)–(4). Mean relative residual $\langle (m_{\text{pred}}^2 - m_{\text{lat}}^2)/m_{\text{lat}}^2 \rangle$ on AT2021 lattice data, 78 channels (excluding 5 ditorelon-dominated 2^{+-} channels). $|\cdot|$ denotes the absolute residual mean.

J^P block	$\langle \delta m^2/m^2 \rangle$	$ \cdot $	n
$J = 0$	−0.132	0.119	20
$J = 1$	−0.066	0.137	20
$J = 2$	−0.069	0.118	25
$J = 3$	−0.135	0.154	9
$J = 4$	0.000	0.078	4
overall	−0.082	0.140	78

the limit of well-separated channels. The excitation level $\text{ex} + \xi_\star$ shows the same heat-kernel offset $\xi_\star$ as the Casimir spine, suggesting a single underlying short-distance exponent.

Universal C -splitting and the $\eta_\infty = 1/2$ discovery

New finding (this work). Channels with $C = -1$ are systematically heavier than their $C = +1$ counterparts. The empirical splitting ratio $\eta(N) \equiv m_{C=-1}^2/m_{C=+1}^2 - 1$ obeys, to within 2.8%,

$$\eta_\infty = \frac{1}{2} \quad (\text{large-}N \text{ limit, NEW}),$$

and the leading $1/N^2$ correction is fitted, to within 0.2%, by

$$c_\eta = -\beta/3 = -16/21 \quad (\text{NEW}).$$

Combining these, $\eta(N) = 1/2 - \beta/(3N^2)$ is now fully determined by rationals that already appear elsewhere in the formula. The factor of 3 in $\beta/(3N^2)$ is naturally read as the number of transverse spatial dimensions ($d_\perp = 3$ in 4D); the structural identity $c_\eta = -\beta/3$ then links the parity and conjugation sectors via a single rational.

CROSS- N EMPIRICAL VERIFICATION

We test the closed form (3)–(4) against the Athenodorou–Teper N -scan [2], which covers $SU(N)$ for $N \in \{3, 4, 5, 6, 8\}$ on a total of ~ 83 channels grouped by (J, P, C, ex) . After excluding 5 channels of 2^{+-} type that are ditorelon-dominated (multi-glueball scattering states, see point (L3) below), we are left with 78 channels.

Table I summarizes the results. The mean residual is 14.0% in absolute value, comparable to the typical statistical-plus-systematic errors of the AT2021 large- N continuum extrapolation (the published error bars on individual masses are in the 3–10% range, with cross-channel correlations). The $J = 3$ block is mildly under-

predicted (-13.5%), consistent with its proximity to the $0^{++} \otimes 2^{++}$ scattering threshold (point (L3) below).

A fully-free 4-parameter fit (varying $\alpha, \beta, \gamma, \eta$) yields 12.9% , only 1.1% better than the all-rational version above. The parameter-free formula is therefore essentially as good as the best empirical fit on the same data.

DISCOVERIES $\eta_\infty = 1/2$ AND $c_\eta = -\beta/3$

The C -splitting law $\eta(N)$ was extracted as follows. For each N , we fit $\eta(N)$ as a free scalar against the residual C -asymmetry of (3) with $\Phi_C \equiv 1$, giving the five values

$$\eta(3) = 0.405, \quad \eta(4) = 0.409, \quad \eta(5) = 0.487,$$

$$\eta(6) = 0.485, \quad \eta(8) = 0.448.$$

A one-parameter functional fit $\eta(N) = a + b/N^2$ returns $a = 0.486(\pm 0.014)$, $b = -0.764(\pm 0.04)$, both within $2-3\%$ of the rationals $a = 1/2$ and $b = -16/21$. The fit $\eta(N) = 1/2 - 16/(21N^2)$ predicts

$$\eta(6) \simeq 0.479 \quad \text{vs} \quad 0.485 \text{ measured,}$$

i.e. 1.3% agreement on the most precisely measured AT2021 point.

Interpretation. The asymptotic value $\eta_\infty = 1/2$ is the structural large- N statement *the $C = -1$ tube is on average 50% heavier than the $C = +1$ tube*. We do not commit to a mechanism here; candidate routes include (i) spin- $1/2$ weight in the adjoint representation, (ii) axial $\mathbb{Z}_2$ anomaly, and (iii) the ratio of complex to real representation dimensions.

The $-\beta/3$ link. That the leading $1/N^2$ correction is exactly $-\beta/3$ ties the C -sector to the parity sector through a single rational. The factor of 3 matches the three transverse dimensions of the flux tube, hinting at a geometric (rather than dynamical) origin.

HONEST LIMITATIONS

(L1) Absolute scale. The string tension $\sqrt{\sigma_0}$ enters as a dimensional input. We do not predict it from arithmetic; the no-go theorems of [19] (in particular T1 of that work, on dimensional transmutation) make this explicit.

(L2) 14% residual. The mean absolute residual on 78 filtered channels is $\sim 14\%$. This is comparable to the published AT2021 systematic-plus-statistical errors on individual masses and on the large- N continuum extrapolation, but is by no means competitive with a precision lattice calculation. Lüscher finite-volume corrections [15, 16], applied on top of (3), recover at best 5% further on our data, suggesting that AT2021 published values are

TABLE II. Falsifiable predictions for selected channels on the $SU(N)$ lattice. Predictions assume $\sqrt{\sigma_0} = 440$ MeV for the $SU(3)$ entries.

Quantity	Predicted	Test
$m(1^{++})_{SU(3)}$	$\simeq 2501$ MeV	new measurement
$m(6^{++})_{SU(3)}$	$\simeq 6.0\sqrt{\sigma_0} \cdot J$ (Regge)	new measurement
$\eta_{SU(6)}$	0.479	0.485 [2]
$\eta_{SU(\infty)}$	$1/2$ exactly	$SU(10)$ extrapolation

already extrapolated to infinite volume to within a few percent.

(L3) The 2^{+-} channels. Five 2^{+-} channels (one per N in our sample) are systematically over-predicted by $50-60\%$ in the closed form. These are known to be ditorelon-dominated multi-glueball scattering states [2], mixing into the lowest single-particle eigenvalue from above. The formula (3)–(4) is therefore restricted to channels that are not contaminated by such low-lying scattering states.

(L4) Cross-group test pending. Bennett *et al.* [3] have published an analogous lattice spectrum for $Sp(2N)$ with $N \in \{1, 2, 3\}$. A preliminary fit with the same K^2 and a group-specific $F_{Sp}(N)$ agrees to 2.5% at the ground state, confirming K as group-universal; the corresponding cross-group test of c^2 and of $\eta(N)$ is ongoing.

(L5) Lean status. Only K , F , and ξ_* are PROVED in Lean 4. The spectral factor c^2 and the discoveries $\eta_\infty = 1/2$, $c_\eta = -\beta/3$ are empirical (TIER 3, SUPPORTED).

FALSIFIABLE PREDICTIONS

The closed form (3)–(4) furnishes parameter-free predictions for as-yet-unmeasured channels. Three of them have been singled out below for near-term lattice falsification.

A measurement deviating from any of these by more than 5σ would falsify the corresponding sector of (3). The $\eta_{SU(6)}$ prediction is the sharpest available falsifier today: the closed form gives 0.479 , the published AT2021 value is 0.485 , in agreement at the 1.3% level (consistent with the AT2021 errors). A future $SU(10)$ data set, by contrast, would distinguish $1/2$ from competing fits.

CONCLUSION AND ROADMAP

We have written down a closed form for the mass-gap ratio m^2/σ_0 of 4D pure $SU(N)$ Yang–Mills in every $J^{PC}(\text{ex})$ channel at every $N \geq 3$, with all coefficients rational and the universal transcendental $\sqrt{4\pi e/3}$ as the only non-rational ingredient. The dimensional scale enters via $\sqrt{\sigma_0}$. On the 78-channel AT2021 sample the

formula matches the lattice data at the 14% level, comparable to the published lattice systematics.

The roadmap toward the Clay problem is as follows. (i) Independent $\mathrm{Sp}(2N)$ tests [3] of $\eta(N)$, with the same two rationals $\eta_\infty = 1/2$ and $c_\eta = -\beta/3$. (ii) A Lüscher-style V -correction analysis on a precision $\mathrm{SU}(3)$ lattice to pin down the residual 5%. (iii) Lean-4 mechanization of the spectral factor c^2 , extending the existing $K, F, \xi_\star$ kernel. (iv) An arithmetic derivation of $\beta = 16/7$ from a parity-genus identity, currently the most opaque rational in (3)–(4).

Acknowledgements. The author thanks the `crossed-cosmos` open notebook project for the lattice data tabulations used in the cross- N verification. All numerical work was carried out in `PARI/GP 2.15`, with cross-checks in `Python`. The Lean 4 proofs of $K, F, \xi_\star$ are kernel-verified at commit `Crossed/Transport.lean`.

Data availability. Lattice data are from [2, 3]; worksheets are at the project repository.

* kevin.remondriere@gmail.com

- [1] A. Jaffe and E. Witten, *Quantum Yang–Mills theory*, Clay Mathematics Institute Millennium Problem description (2000).
- [2] A. Athenodorou and M. Teper, *SU(N) gauge theories in 3 + 1 dimensions: glueball spectra, string tensions and topology*, JHEP **12** (2021) 082, [arXiv:2106.00364](#).
- [3] E. Bennett *et al.*, *Sp(2N) lattice gauge theories and extensions of the Standard Model of particle physics*, EPJ Plus **139** (2024); [arXiv:2103.00665](#).
- [4] C. J. Morningstar and M. J. Peardon, *The glueball spectrum from an anisotropic lattice study*, Phys. Rev. D **60** (1999) 034509; [arXiv:hep-lat/9901004](#).
- [5] B. Lucini, M. Teper, and U. Wenger, *Glueballs and k-strings in SU(N) gauge theories: calculations with improved operators*, JHEP **06** (2004) 012; [arXiv:hep-lat/0404008](#).
- [6] E. T. Jaynes, *Information theory and statistical mechanics*, Phys. Rev. **106** (1957) 620.
- [7] E. T. Jaynes, *Information theory and statistical mechanics. II.*, Phys. Rev. **108** (1957) 171.
- [8] J. Elstrodt, F. Grunewald, and J. Mennicke, *Groups acting on hyperbolic space: harmonic analysis and number theory*, Springer Monographs in Mathematics (1998).
- [9] P. Sarnak, *The arithmetic and geometry of some hyperbolic three-manifolds*, Acta Math. **151** (1983) 253–295.
- [10] K. Rémondrière, *A topological invariant of Bianchi 3-orbifolds from the identity term of the Selberg pretrace formula*, preprint, Comptes Rendus de l’Académie des Sciences (May 2026).
- [11] G. ’t Hooft, *A planar diagram theory for strong interactions*, Nucl. Phys. B **72** (1974) 461.
- [12] R. Dijkgraaf and E. Witten, *Topological gauge theories and group cohomology*, Commun. Math. Phys. **129** (1990) 393–429.
- [13] M. Lüscher and P. Weisz, *Quark confinement and the bosonic string*, JHEP **07** (2002) 049; [arXiv:hep-lat/0207003](#).
- [14] M. Lüscher and P. Weisz, *String excitation energies in SU(N) gauge theories beyond the free-string approximation*, JHEP **07** (2004) 014; [arXiv:hep-th/0406205](#).
- [15] M. Lüscher, *Volume dependence of the energy spectrum in massive quantum field theories. I. Stable particle states*, Commun. Math. Phys. **104** (1986) 177.
- [16] M. Lüscher, *Volume dependence of the energy spectrum in massive quantum field theories. II. Scattering states*, Commun. Math. Phys. **105** (1986) 153.
- [17] J. M. Maldacena, *The large-N limit of superconformal field theories and supergravity*, Adv. Theor. Math. Phys. **2** (1998) 231–252; [arXiv:hep-th/9711200](#).
- [18] A. Chandra, I. Chevyrev, M. Hairer, and H. Shen, *Stochastic quantisation of Yang–Mills–Higgs in 3D*, Invent. Math. (2024); [arXiv:2201.03487](#).
- [19] K. Rémondrière, *No-go theorems for the arithmetic determination of the Yang–Mills mass gap absolute scale*, preprint, Physical Review Letters submission (May 2026).